

Professional Skills Matrix and Action Plan

Ariel Mella · PCOM7E Machine Learning · April 2026 intake · University of Essex Online

Skills Matrix (state at end of module)

Skill levels: **Aware** (basic understanding) · **Trained** (applies independently in defined contexts) · **Proficient** (broad and in-depth) · **Expert** (subject-matter expert, leads and trains others).

Skill area	Level	Evidence
1. Commercial Awareness	Expert	20+ years IT industry leadership across IBM, VMware and Broadcom; cyber-resiliency and cloud portfolio ownership.
2. Time Management	Expert	Track 1 submitted two days ahead of the deadline; sustained weekly cadence across 12 units.
3. Critical Thinking and Analysis	Expert	Three-tier design framework authored for Track 1; variable-change probe on every tutorial notebook.
4. Communication and Literacy	Expert	1,000-word executive report; CD1 cross-sector synthesis; commercial sales and pre-sales background.
5. IT and Digital	Expert	IT Architect across enterprise estates; this e-portfolio on Jekyll and GitHub Pages; five-script Python pipeline.
6. Numeracy	Proficient	Regression, clustering and metrics (R^2 , ROC AUC, F1, RMSE) across Units 3-11; MBA quantitative methods.
7. Research	Proficient	Harvard Cite Them Right across ~140 references; literature-anchored cleaning rules; MBA dissertation.
8. Interpersonal	Expert	Group D Coordinator role; pre-sales and sales leadership; multinational distributed teams.
9. Problem-solving	Expert	20+ years architect; caught five own-work flaws on the individual presentation before submission.
10. Ethical Awareness	Proficient	Dual-use CNN argument across Units 9 and 10; GDPR-aware peer redaction across the e-portfolio.
11. Teamwork and Leadership	Expert	Group D Coordinator; engagement-based task allocation (Weick, 1976); 20+ years leading distributed teams.
12. Critical Reflection	Proficient	1,000-word reflective piece (Rolfe et al., 2001); 3 W's reflection across 17 activity cards.
13. Classical ML - Regression (Units 3, 4)	Trained	Linear, multiple and polynomial regression; Track 1 log-price model.
14. Classical ML - Clustering and Similarity (Units 5, 6)	Trained	Jaccard, K-Means with elbow / silhouette / ARI; Track 1 segmentation.
15. Deep Learning - Perceptron and MLP (Unit 7)	Trained	Simple perceptron; AND in 6 epochs; 2-3-1 sigmoid XOR by back-propagation.
16. Deep Learning - CNN (Units 9, 10, 11)	Trained	CIFAR-10 trained to 77.82% test accuracy; CNN Explainer 4-case study (Wang et al., 2020); pneumonia ML-vs-DL individual presentation.
17. ML Evaluation and Metrics (Unit 11)	Trained	Confusion matrix, F1, ROC AUC, MAE / RMSE / R^2 ; parameter-change exercises on AUC and polynomial degree.
18. Explainability (Units 8, 10)	Aware	SHAP-style decomposition on a Matrix-Factorisation prediction; CNN Explainer pipeline walk-through.

Action Plan (3 to 12 months post-module)

The Action Plan responds to the Skills Matrix above. It targets the rows where I sit at Aware or Trained and the Proficient rows where I want to push toward Expert. The plan is intentionally tight: six development goals with concrete actions, evidence and timelines.

Skill area	Goal	Action	Timeline
Deep Learning	Trained → Proficient	Self-led CNN project; read Goodfellow, Bengio and Courville (2016) end-to-end.	6 months
Python production ML	Trained → Proficient	Deploy a small ML service with CI/CD on a personal project.	6 months
Explainability	Aware → Trained	Apply SHAP to one deployed model; produce one written case study.	3 months
Ethical Awareness	Proficient → Expert	One position paper on dual-use CNN deployment; monthly EU AI Act tracking.	12 months
Research	Proficient → Expert	One peer-reviewable piece connecting cybersecurity and AI.	12 months
Critical Reflection	Proficient → Expert	Monthly reflective journal post-MSc; three-tier design framework applied to one professional decision document per quarter.	Ongoing

Reviewed and committed by Ariel Mella, 14 May 2026. Next review: end of MSc programme.